

Ans 1. Arteries are the thick-walled blood vessels that carry oxygenated blood away from the heart to various body tissues, with the only exception being the pulmonary artery.

Ans 2. Chloroplast is the organelle primarily involved in photosynthesis. It contains green pigment chlorophyll which absorbs solar energy.

Ans 3. **Role of Chloroplasts in Photosynthesis:**

- Main pigments: Chlorophyll a, Chlorophyll b, and Carotenoids.

1. Light Reaction: Takes place in thylakoid membranes. Solar energy is absorbed to split water (photolysis) releasing O₂, ATP & NADPH.

2. Dark Reaction (Calvin Cycle): Takes place in stroma. Uses ATP & NADPH from light reaction to fix CO₂ into glucose.

Ans 5(a) Transpiration is the loss of water in the form of vapor from the aerial parts of the plant, especially stomata in leaves. It creates transpiration pull that helps pull water from roots up to leaves.

Ans 5(b) Two environmental factors increasing transpiration rate:

1. Higher Temperature: Increases kinetic energy & evaporation rate.

2. High Wind Velocity: Sweeps away humid air around leaves faster.